

**1E3106**

Roll No. ,

Total No. of Pages: **4****1E3106****B. Tech. I - Sem. (Main/Back/Re-back) Exam., Jan. - 2026**  
**1FY3-06 Programming for Problem Solving****Time: 3 Hours****Maximum Marks: 70***Instructions to Candidates:**Attempt all ten questions from Part A, five questions out of seven questions from Part B and three questions out of five questions from Part C.**Schematic diagrams must be shown wherever necessary. Any data you feel missing may suitably be assumed and stated clearly. Units of quantities used /calculated must be stated clearly.**Use of following supporting material is permitted during examination. (Mentioned in form No. 205)*1. NIL2. NIL**PART - A****[10×2=20]****(Answer should be given up to 25 words only)****All questions are compulsory**

- Q.1 Differentiate between compiler and interpreter.
- Q.2 Find out 9's complement of this number:  $(744)_{10}$
- Q.3 What is secondary memory? Give examples.
- Q.4 What is the role of cache memory in computer performance?
- Q.5 Define the role of the Control Unit (CU) in the processing operations of a computer system.

- Q.6 What are different types of computer languages?
- Q.7 Compare different types of computer memories based on various criteria like storage capacity, access speed and cost per bit.
- Q.8 Define the properties of C arrays.
- Q.9 What do you understand by the term type casting in C programming?
- Q.10 What are different bitwise operators in C?

### **PART – B**

**[5×4=20]**

**(Analytical/Problem solving questions)**

**Attempt any five questions**

- Q.1 Illustrate the purpose and syntax of various decision-making statements in C.
- Q.2 Convert the following -
- (i)  $(345)_{10} = (?)_2$
  - (ii)  $(23B)_{16} = (?)_{10}$
  - (iii)  $(3815)_{10} = (?)_8$
  - (iv)  $(1111)_2 = (?)_{10}$
- Q.3 Describe the representation of alphabets using ASCII and Unicode.
- Q.4 Explain in detail the difference between SRAM (Static RAM) and DRAM (Dynamic RAM). Describe how each type stores data, their speed, cost, power consumption and common applications.
- Q.5 Explain one-dimensional and two-dimensional arrays in C with suitable examples. Also, write a C program to find the sum of all elements in a one-dimensional array.
- Q.6 Explain the Von Neumann architecture in detail. Discuss the steps involved in the instruction execution cycle and highlight the limitations of this architecture.

Q.7 Write a C program to print pattern -

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* * * * *
* * * *
* * * *
* * *
* * *
*
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## **PART - C**

[3×10=30]

**(Descriptive/Analytical/Problem Solving/Design Questions)**

**Attempt any three questions**

- Q.1 Write and explain the logic of a C program that reads a large text file line by line, counts the frequency of each word and stores the results in another output file.
- Q.2 What are different types of C functions? Find the factorial of a number using recursion. What are call by value and call by reference?
- Q.3 Write a C program to create a student structure using name, id and average marks using proper data types, using array of structures store and print five students' data.
- Q.4 What are pointer variables? What different applications of pointers also create a C program to demonstrate array of pointer?
- Q.5 Write a C program to perform matrix multiplication also design algorithm and flowchart with neat diagram and proper flowchart symbols.
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